SVM https://chatgpt.com/share/69ea646e-5878-8322-afc8-a59cf5e89f99

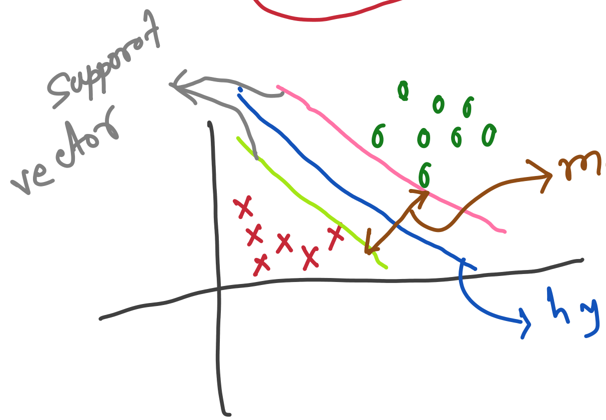
Wednesday, April 22, 2026 12:13 PM

goal: maximum the margin

SVM → supervised Algorithm

SVM → make margin → (i) hard margin
(ii) Soft "

SVM → to Separate data



margin → (i) margin বড় → hard margin
(ii) " ছোট → soft margin

hyperline

(i) Hard margin → used for good dataset
↳ data must be perfect Separable

(ii) Soft " → used for noisy data
↳ outlier থাকে।
↳ use value of c → c <<< তো ভালো
→ c >>> " খারাপ

When Use SVM :-

(i) Data is clear/Separable :→ linear SVM

(ii) Data is mixed → no exact shape
↳ Decision Tree
↳ Random forest

(iii) Mixed but curved shape
↳ non-linear SVM
↳ kernel Trick

(iv) Very large data
↳ million of rows
↳ logistic Regression
↳ SGD classifier

👉
Model selection depends on data characteristics. For linear data we use Logistic Regression or Linear SVM. For non-linear structured data we use SVM with kernel or Decision Tree. For noisy data we prefer simpler models. For large datasets SGDClassifier is efficient, and for interpretability Decision Tree is preferred.